

MOUNIKA TULASI

Data Analyst

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SUMMARY

Data Analyst with 3 years' experience in **financial services, healthcare, and consulting**, skilled in **Python, SQL, Snowflake, Apache Airflow, AWS Glue, Kafka, Power BI, and Tableau** to build scalable **pipelines** and deliver actionable **insights**. Driving results through a **5-day faster month-end close**, improved **data quality** across **4 multi-national audits**, and **\$800K in operational savings**. Applied **analytical models** and **visualization best practices** with **cross-functional teams, architecture planning, and KPI standardization** to enhance **executive dashboard adoption**.

EXPERIENCE

Data Analyst | BNY Financial | USA

Mar 2025 – Present

- Engineered a fully ETL pipeline using **Apache Airflow, AWS Glue, and PostgreSQL** to consolidate feeds from 5 legacy systems into **Snowflake**, cutting month-end close from **10 to 5 business days**.
- Orchestrated advanced **SQL** (window functions, CTEs, query optimization) and **Python Pandas/PySpark** transformations on 48M+ transactional records, aligning outputs with compliance rules to remove **320 reconciliation anomalies annually**.
- Synthesized streaming and batch datasets via **Kafka** with **statistical modeling** in **R** and **Power BI** to detect revenue variance patterns, enabling leadership to reallocate **\$1.2M** toward high-margin portfolios.
- Directed **schema architecture** for a **dbt-powered semantic layer** in **Snowflake**, enabling unified **KPI logic** for **7 finance departments** and boosting **dashboard reliability** in **Tableau** and **Power BI**.

Data Analyst | McKesson | USA

Mar 2024 – Feb 2025

- Constructed **data integration workflows** with **AWS Glue, Informatica, and SSIS** to migrate **9.2M pharmaceutical transactions** into **Snowflake**, eliminating **36 hours** of monthly manual preparation.
- Optimized complex **SQL** queries and **Python Pandas/PySpark** scripts to process 42M+ clinical trial data points, cutting query execution time from **12 minutes to under 4 minutes**.
- Crafted **Tableau and Power BI** dashboards for **continuous tracking of 18 clinical KPIs**, reducing trial phases by **5 weeks**.
- Integrated **Kafka streaming data** into **Snowflake** and enforced **validation** with **Great Expectations**, ensuring **dataset integrity** across **6 compliance audits**.
- Analyzed patient safety datasets using **R** and statistical analysis methods to identify adverse event trends, supporting the reallocation of **\$640K** in research budgets toward higher-risk studies.
- Coordinated **Agile sprint cycles** with **BI, QA, and clinical stakeholders** to define **technical requirements** for **4 integrated reporting pipelines**, improving delivery timelines for **regulatory submissions**.

Data Analyst Intern | KPMG | India

Jun 2020 – Nov 2021

- Engineered **ETL pipelines** with **Informatica, SSIS, and Python Pandas** to unify **3.9M tax compliance records** in **Snowflake**, shortening audit processing by **5 business days**.
- Created **Power BI** and **Tableau** dashboards to visualize **18 client KPIs**, enabling senior consultants to present findings that accelerated decision-making for corporate audits covering **₹180M** in transactions.
- Optimized **SQL** queries and **PySpark** scripts to process **21.5M financial transactions**, reducing execution time from **8 minutes to under 3 minutes** while maintaining compliance with audit protocols.
- Simulated risk assessment scenarios in **R** to identify anomalies in procurement data, enabling the recovery of **₹14.6M** in overbilled vendor payments across 2 client accounts.
- Facilitated **Agile backlog refinement sessions** with **BI developers and compliance teams**, setting **deliverables** for **3 reporting pipelines** that accelerated statutory **audit readiness** by **7 business days**.
- Led the creation of a **dbt-modeled semantic structure** in **Snowflake**, aligning **KPI definitions** across **4 audit teams** to ensure consistency in **Tableau** executive reporting.

TECHNICAL SKILLS

Big Data & Distributed Systems: Apache Kafka, Apache Hive, Apache Airflow, Apache Spark, Hadoop

Programming & Scripting: Python, SQL (Window Functions, CTEs, Query Optimization), R, HTML, Spark SQL, C

Databases & Data Warehousing: MySQL, Oracle, MongoDB, SQL Server, PostgreSQL, Amazon Redshift, Snowflake, Dimensional Modeling (Star/Snowflake Schema)

Cloud & ETL Tools: AWS (Glue, S3, Redshift, Lambda, IAM), Informatica, SSIS, Azure Data Factory, GCP BigQuery

Data Processing & Analytics: Pandas, PySpark, Data Cleaning, Data Transformation, Data Modeling, Data Quality Checks

Data Visualization & BI: Tableau, Power BI

Workflow Orchestration & Automation: Apache Airflow, dbt (Data Build Tool)

Containerization & Deployment: Docker, Kubernetes

Version Control & DevOps: Git, GitHub, Bitbucket, CI/CD

Methodologies: Agile (Scrum, Kanban)

PROJECTS

AI-Driven Hospital Readmission Reduction System

- Built a **Python** machine learning model with **scikit-learn** and **XGBoost** on **1.8M hospital records** in **Snowflake** and **PostgreSQL**, predicting high-risk patients and reducing readmissions by **~410 cases annually**.
- Streamlined EHR, lab, and discharge data ingestion using **Apache Airflow** and **AWS Glue**, cutting preprocessing time from **75 to 18 minutes** for care coordination teams.

AI-Powered Supply Chain Disruption Forecasting Tool

- Developed a forecasting model in **PySpark** and **R** using shipment data, weather patterns, and supplier metrics from **Snowflake** and **Kafka**, providing early warnings to prevent **\$860K in losses annually**.
- Created **Power BI** dashboards with real-time alerts and **LangChain**-based natural language querying, reducing disruption identification time from **1 hour to 15 minutes**.

EDUCATION

Master's in Computer Science | Oklahoma City University | USA

Jan 2021 – Dec 2023

B.Tech in Computer Science Engineering | Vignan's Foundation for Science, Technology & Research | India

Aug 2017 – Apr 2021